THE I-ADIC HOMOLOGY OF A COMPLEX OF FINITE MODULES IS PRO-ISOMORPHIC TO THE POWERS OF ITS HOMOLOGY

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The Artin-Rees lemma for an ideal in a Noetherian ring. applied at proof lines 29466-29470 through the cited Algebra lemmas. Supplies the two constants c_1 and c_2 whose sum controls the comparison maps.
- Pro-objects of modules and morphisms between them. the final conclusion at statement lines 29455-29456. Fixes the sense in which (H_n) and (I^nH) are isomorphic.

Source reference. Original lemma 00IN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 685–685. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00IO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 686–686. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let I be an ideal of a Noetherian ring A. Let $K \xrightarrow{\alpha} L \xrightarrow{\beta} M$ be a complex of finite A-modules. Set $H = \mathrm{Ker}(\beta)/\mathrm{Im}(\alpha)$. For $n \geq 0$ let*

$$I^n K \xrightarrow{\alpha_n} I^n L \xrightarrow{\beta_n} I^n M$$

be the induced complex. Set $H_n = \mathrm{Ker}(\beta_n)/\mathrm{Im}(\alpha_n)$. Then there are canonical A-module maps

$$\ldots \to H_3 \to H_2 \to H_1 \to H$$

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There exists a $c > 0$ such that for $n \geq c$ the image of $H_n \to H$ is contained in $I^{n-c}H$ and there is a canonical A-module map $I^nH \to H_{n-c}$ such that the compositions

$$I^nH \to H_{n-c} \to I^{n-2c}H \quad \text{and} \quad H_n \to I^{n-c}H \to H_{n-2c}$$

are the canonical ones. In particular, the inverse systems (H_n) and (I^nH) are isomorphic as pro-objects of Mod_A.

Proof. We have $H_n = \mathrm{Ker}(\beta) \cap I^nL/\alpha(I^nK)$. Since $\mathrm{Ker}(\beta) \cap I^nL \subset \mathrm{Ker}(\beta) \cap I^{n-1}L$ and $\alpha(I^nK) \subset \alpha(I^{n-1}K)$ we get the maps $H_n \to H_{n-1}$. Similarly for the map $H_1 \to H$.

By Artin-Rees we may choose $c_1, c_2 \geq 0$ such that $\mathrm{Im}(\alpha) \cap I^nL \subset \alpha(I^{n-c_1}K)$ for $n \geq c_1$ and $\mathrm{Ker}(\beta) \cap I^nL \subset I^{n-c_2}\mathrm{Ker}(\beta)$ for $n \geq c_2$, see Algebra, Lemmas 00IO and 00IN. Set $c = c_1 + c_2$.

It follows immediately from our choice of $c \geq c_2$ that for $n \geq c$ the image of $H_n \to H$ is contained in $I^{n-c}H$.

Let $n \geq c$. We define $\psi_n : I^nH \to H_{n-c}$ as follows. Say $x \in I^nH$. Choose $y \in I^n\mathrm{Ker}(\beta)$ representing x. We set $\psi_n(x)$ equal to the class of y in H_{n-c}. To see this is well defined, suppose we have a second choice y' as above for x. Then $y' - y \in \mathrm{Im}(\alpha)$. By our choice of $c \geq c_1$ we conclude that $y' - y \in \alpha(I^{n-c}K)$ which implies that y and y' represent the same element of H_{n-c}. Thus ψ_n is well defined.

The statements on the compositions $I^nH \to H_{n-c} \to I^{n-2c}H$ and $H_n \to I^{n-c}H \to H_{n-2c}$ follow immediately from our definitions. □